

Akshay Ashok Nair

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EDUCATION

University of Washington

Bachelor of Science in Electrical and Computer Engineering

GPA: 3.7 | Relevant Coursework: Digital Circuits and Logic Design, Computer Architecture, Circuit Analysis, Signal Processing, Data Structures

Seattle, WA

Expected June 2028

TECHNICAL SKILLS

Languages: SystemVerilog, Verilog, ARM Assembly, C, C++, Python, Java, JavaScript, C#

Tools: Altium Designer, Vivado, ModelSim, Xilinx Artix-7 FPGA, Arduino IDE, Jetson Nano, Git, Linux, PyTorch, TensorFlow, Keras, OpenCV, ROS

Areas: PCB Design, Schematic Capture, PCB Layout & Routing, RTL Design, Digital Logic Design, Computer Architecture, Pipelining, Hazard Forwarding, FSMs, Testbenches, Functional Verification, Timing Analysis, Embedded Systems, I2C, USB, Signal Processing, Deep Learning, Computer Vision

PROJECTS

FPGA CNN Accelerator for Digit Recognition | [SystemVerilog, Vivado, Python, FPGA] June 2026 – Present

- Trained a CNN on MNIST (**98%** float32 accuracy), then quantized weights to **int8** with only 0.05 point accuracy loss
- Built a SystemVerilog inference pipeline (2 conv, 2 max-pool, FC, argmax); verified **RTL** against **Python** reference outputs
- Fixed **MAC** overflow and timing paths via calibrated requantization and kernel prefetch, raising accuracy **30%** to **100%**

5-Stage Pipelined ARM LEGv8 CPU | [SystemVerilog, ModelSim] June 2026 – Present

- Built a **gate-level** 5-stage pipelined **ARM LEGv8** CPU (IF/ID/EX/MEM/WB) with an ALU, adders, and register file
- Designed a 3-source priority-encoded **forwarding** network (EX/MEM/WB) resolved early in ID, removing all stalls
- Resolved branches/CBZ/BR early via a **delay-slot** model plus a flush mode; verified in 9 **ModelSim** assembly tests

Autonomous LiDAR Maze Navigation | [C++, SystemVerilog, Arduino, FPGA, LiDAR] Mar. 2025 – Present

- Building an autonomous vehicle that fuses **LiDAR** point-cloud sensing with a wall-following algorithm to solve a maze
- Programming **FPGA** control logic in **Verilog** and Arduino firmware for sensor/motor integration, targeting sub-30s runs

Real-Time Face Recognition System | [Python, OpenCV, PyTorch, Jetson Nano] Sept. 2025 – Dec. 2025

- Deployed a real-time face recognition model on a **Jetson Nano** with a live **OpenCV** camera-input pipeline
- Optimized the architecture for **edge deployment**, hitting **95%** accuracy across 50 member faces under compute constraints

STM32F411 4-Layer PCB | [Altium Designer] Aug. 2026

- Completed a **4-layer STM32F411** board build in Altium: schematic capture, placement, and routing
- Routed a 90-ohm impedance-controlled **USB** pair plus **I2C**, **SWD**, and LDO power nets over inner ground planes
- Generated **Gerbers**, pick-and-place, and **BOM** outputs for JLCPCB assembly after clearing design-rule checks (**DRC**)

EXPERIENCE

Machine Learning Intern - Foy Lab Jan. 2026 – Present

University of Washington

Seattle, WA

- Trained a CellCNN-style convolutional neural network on **2,000+** patient samples (**8,192** cells each) to predict ER status
- Ensured reproducibility with stratified train/val/test splits, z-score normalization, and parameter-encoded caching
- Built a dual-backend pipeline feeding a shared data loader into a **Keras CNN** and a **CytoDx R** model, AUROC **0.73**

Project Lead - The Boring Club (Face Recognition System, LiDAR Maze Navigation) Sept. 2025 – Present

University of Washington

Seattle, WA

- Directed **15** student engineers across two hardware/ML projects, defining system architecture and weekly milestones
- Drove design trade-offs for edge deployment on constrained hardware, achieving **95%** accuracy on both systems

Machine Learning Researcher June 2021 – June 2024

Burkitt's Lymphoma Fund for Africa

Seattle, WA

- Developed and fine-tuned **CNNs** in **PyTorch** for Burkitt's Lymphoma detection on histopathology slides, **90%+** AUROC
- Partnered with UW, Fred Hutch, and Phenopath Labs on system design and validation, yielding a first-author publication

Research Assistant - Sensors, Energy, and Automation Lab Apr. 2024 – Sept. 2025

University of Washington

Seattle, WA

- Built **JavaScript** and **Python/Selenium** automation tools to streamline member tracking and research data workflows
- Prepared a research manuscript with graduate researchers for submission to the **CDC**

Software Development Intern June 2023 – Sept. 2023

Liberated Logic

Bothell, WA

- Prototyped a user-centered website with interactive Figma wireframes and **React** components to improve early usability

PUBLICATIONS

Nair, A., et al. *An AI based, open access screening tool for early diagnosis of Burkitt Lymphoma*. Frontiers in Medicine, 2024.